

SAHANI RAJAPAKSHE

Software Engineer | Full Stack & AI Solutions

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https://sahani-portfolio.vercel.app/

PROFESSIONAL SUMMARY

AI-focused Full Stack Software Engineer with 4+ years of experience building scalable enterprise applications, AI-powered products, intelligent assistants, semantic search solutions, and distributed backend services across fintech, healthcare, and aviation-tech domains. Skilled in ReactJS, Node.js, .NET, Java Spring Boot, PostgreSQL (pgvector), LangChain, OpenAI APIs, REST APIs, Microservices Architecture, OAuth2/JWT authentication, Redis caching, and CI/CD pipelines. Experienced in AI copilots, RAG-based systems, semantic search, vector embeddings, AI-driven decision support, prompt engineering, and production-grade backend engineering.

TECHNICAL SKILLS

- **Programming Languages:** TypeScript, JavaScript, C#, SQL
- **Frontend Technologies:** ReactJS, Next.js, Material UI (MUI), HTML, CSS, Tailwind CSS
- **Backend & API Development:** NestJS, ASP.NET Core, .NET, Spring Boot, REST APIs, GraphQL, Microservices
- **Databases & Data Systems:** PostgreSQL, MySQL, SQL Server, Redis
- **Generative AI & LLMs:** OpenAI API, RAG, LangChain, Semantic Search, Vector Embeddings, Prompt Engineering, LLM Integration
- **Cloud & DevOps:** Azure, Docker, GitHub Actions, CI/CD
- **Architecture & Engineering Practices:** OAuth2, JWT, RBAC, Distributed Systems, Agile/Scrum
- **AI-assisted development tools :** ChatGPT, GitHub Copilot, Cursor, Claude)

EXPERIENCE

Full Stack Developer | Mediio.net | South Korea(Remote) | Oct 2025 – Present

- Designed and implemented AI-powered semantic search, AI recommendation systems using OpenAI, vector embeddings and PostgreSQL (pgvector) to improve information retrieval accuracy for the Medical App.
- Designed and developed an AI-powered product enrichment platform that automatically extracted product information from website URLs and leveraged LLMs to generate structured product descriptions, specifications, SEO-optimized content, product metadata, and AI-generated product images, supported by scalable backend APIs and content processing pipelines for automated catalog creation and enrichment for the healthcare system.
- Built AI Insight feature analyzing structured and unstructured datasets to generate decision-support summaries.
- Developed AI chatbot assistant using OpenAI, LLM and RAG-based retrieval systems.
- Developed AI medical assistant using OpenAI, LLM and RAG-based retrieval systems to identify doctor-patient conversations and generate medical reports from discussions and clinical notes
- Developed scalable microservices using Node.js (NestJS) and Java-based services for distributed enterprise applications.
- Implemented secure Identity & Access Management using OAuth2.0, JWT, and RBAC.
- Designed RESTful APIs aligned with OpenAPI standards for integration across services.
- Applied Redis caching and performance optimization strategies to reduce latency.
- Integrated structured logging, metrics, and monitoring to enhance observability of AI and backend services.
- Contributed to CI/CD pipelines (Docker, GitHub Actions) ensuring high-quality, automated deployments.
- Maintained high unit test coverage and participated in production support rotation, resolving defects within SLA.

- Tech Stack: ReactJS, Next.js, TypeScript, Node.js, NestJS, Java, PostgreSQL (pgvector), LLM, RAG, Redis, Docker, GitHub Actions

Associate Software Engineer | Full Stack/.NET | Synapsys LTD, Sri Lanka | Apr 2022 – Sep 2025

- Migrated enterprise banking systems from .NET Framework to .NET Core using Onion Architecture.
 - Designed and implemented high-volume secure REST APIs for financial transaction processing.
 - Optimized complex SQL queries and stored procedures improving transaction throughput by 80%.
 - Implemented secure authentication using JWT and OAuth2.0 standards.
 - Integrated Redis and message queues (RabbitMQ) for distributed reliability.
 - Developed frontend modules using React and JavaScript.
 - Maintained high unit test coverage and participated in CI/CD pipeline automation.
 - Troubleshoot production issues and delivered defect fixes with minimal rework.
 - Collaborated within Agile SDLC and used Jira & Confluence for documentation.
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- Tech Stack: C#, .NET Core (3.1–9), ASP.NET Core, Entity Framework, SQL Server, MySQL, T-SQL, Redis, RabbitMQ, ML.NET, REST APIs, Azure, GitHub Actions, Agile/Scrum

Trainee Software Developer | MobiOs Pvt Ltd, Sri Lanka | Sep 2019 – Mar 2020

- Developed backend REST APIs using Java and SQL Server.
 - Built frontend UI modules using React and JavaScript.
 - Documented APIs and supported integration testing.
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- Tech Stack: Java, SQL Server, REST APIs, React, JavaScript

PERSONAL PROJECTS

AI-Powered Financial Crime Investigation Copilot – [GitHub Link](#) (In Progress)

- Developed a financial crime investigation platform using ASP.NET Core 9, PostgreSQL, CQRS, MediatR, and Clean Architecture.
- Implemented customer, account, and transaction management services with scalable REST APIs and database-driven workflows.
- Designed the foundation for AI-assisted fraud investigation, risk scoring, relationship analysis, and automated investigation report generation.

Secure Auth API with Anomaly Detection – [GitHub Link](#)

- Developed secure REST APIs using .NET 9 with OAuth2.0 and JWT authentication.
- Implemented ML.NET anomaly detection to identify suspicious login activity.
- Integrated CI/CD pipeline for automated build and deployment.

EDUCATION

- BSc. (Hons.) Software Engineering, University of Kelaniya, Sri Lanka (GPA: 3.59/4.0 - Second Upper)
Major: Web Development | Minor: Machine Learning

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate (In Progress)

REFERENCES

References available upon request.